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Probability of PRRS virus detection in pooled processing fluid samples

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ABSTRACT

There has been a tremendous increase in recent years of population-based diagnostic monitoring and surveillance strategies in swine populations. One example is the use of processing fluids (PF) to screen breeding herds for porcine reproductive and respiratory syndrome virus (PRRSV) activity. An important question from practitioners using such methods is on how intensively can the sample be pooled. More specifically, processing fluids of how many litters can be pooled into a single sample for diagnostic testing to preserve a high probability of PRRSV RNA detection at low prevalence situations? The objective of this study was to model the effect of pooling PF samples on the probability of PRRSV RNA detection. For this study, a PRRSV-positive PF field sample with a RT-rtPCR quantification cycle (Cq) value of 28 was selected to represent a litter of 11 pigs with a single viremic piglet. PF samples from a PRRSV-naïve herd were used to perform 6 replications of 8 two-fold serial dilutions of the PRRSV-positive sample, thus modeling the pooling effect (dilution). Each two-fold dilution represented an increase in the number of PRRS-negative pigs in the sample by a factor of 2. Samples were tested for PRRSV RNA by RT-rtPCR and the data was analyzed using linear and probit regression models. There was an average increment of 1.37 points in Ct for each two-fold dilution. The estimated probability of testing positive on RT-rtPCR was 43 %, 80 %, and 95 % when there was a single PRRSv-positive piglet among 784, 492, and 323 PRRSv-negative piglets contributing to the sample respectively. Results from this study support the practice of collecting and aggregating PF samples from multiple litters for PRRSV RNA testing.

1. Introduction

Porcine reproductive and respiratory syndrome virus (PRRSV) continues to be a significant problem in most pork-producing countries. Continuous monitoring of PRRSV shedding is necessary to understand virus dynamics over time and acquire the information necessary to support PRRSV control and elimination programs. Well-designed sampling strategies for PRRSV surveillance must be practical, reliable, and affordable in order to be widely adopted.

By definition, pooling involves intentionally combining individual samples into one, usually in equal proportions. One of the goals of pooling is to reduce the number of tests performed, lowering testing costs while obtaining diagnostic information from a greater proportion of the population (Schisterman and Vexler, 2009). Distinct from pooled samples, "aggregate" samples represent multiple subjects or areas, but in unknown proportions. Typical examples of aggregate samples in veterinary medicine include environmental samples, bulk milk samples, and oral fluids from multiple animals (Rotolo Marisa et al., 2018). Processing fluids are an aggregate sample composed of the serosanguinous exudate derived from the tissues removed during castration or tail docking (i.e., piglet processing), which is typically performed in 2- to 5-day-old piglets (López and Linhares, 2018; Lopez et al., 2018). Following the initial report in 2017, processing fluids have been widely adopted by the swine industry for PRRSV detection (Trevisan et al., 2019). However, questions remain concerning the capabilities and limitations of this approach in PRRSV monitoring and surveillance (Yeske, 2018). In particular, the

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cumulative impact of samples from PRRSV-negative litters on the probability of PRRSV RNA detection in a processing fluid sample by reverse-transcription quantitative polymerase chain reaction (RT-rtPCR) is unknown. Therefore, the goal of this study was to model the probability of PRRSV RNA detection in pooled processing fluids, which in turn can be used for guidelines on the number of litters to include in the aggregate for routine surveillance.

2. Materials and methods

2.1. Study design

To model the effect of diluting one PRRSV RT-rtPCR-positive sample with the addition of PRRSV-negative processing fluids, six replicates of eigth 2-fold serial dilutions (1:2 to 1:512) were produced from a stock solution of PRRSV-positive processing fluid sample (Cq 28) using a composite of PRRSV-free processing fluid samples as the diluent. Serially-diluted samples were tested by RT-rtPCR for the presence of PRRSV RNA and the results were analyzed using regression models to determine the maximum number of piglets that could be included in the sample while assuring the detection of one viremic litter with 95 % confidence.

2.2. Processing fluid samples

Collection of processing fluids was performed in commercial farms by their own staff according to the methods described by López and Linhares (2018) and Lopez et al. (2018). Briefly, a clean plastic container was lined with a disposable plastic bag, and the top was covered with a disposable gauze filter. Tail and testicle tissues were placed on the gauze filter, thereby allowing the fluid to drain into the bag. Fluids were then decanted into a sterile 50 mL plastic container for transportation to the laboratory. Samples were stored in a refrigerator at their farm of origin for up to 3 days and then were shipped to the Research and Development Laboratory of the Iowa State University College of Veterinary Medicine by private courier service. Upon arrival, samples were aliquoted and stored at −20 °C.

One PRRSV RT-rtPCR-positive processing fluid sample (Cq 28) collected from 3- to 5-day-old piglets in a PRRSV positive-unstable (Holtkamp et al., 2011) commercial breeding herd 10 weeks after a PRRSV outbreak was used as the positive stock solution. This Cq value was chosen to represent a processing fluid sample from litters with one viremic piglet based on prior reports (Vilalta et al., 2018; López et al., 2020). The diluent was composed of a collection of processing fluids samples (n = 10) obtained from a PRRSV-naïve (Holtkamp et al., 2011) breeding herd using the same methods described above. There was diagnostic evidence to support the naïve status, including the absence of anti-PRRSV antibodies from 30 serum samples from sows, as well as the absence of PRRSV-associated clinical signs. Moreover, the individual PF utilized as the diluent and the combined solution were individually tested for the presence of PRRSV RNA by RT-rtPCR before and after the experiment to verify the negative status of the samples.

2.3. Diagnostic tests

The nucleic acid extraction procedures were done using the RealPCR* DNA/RNA Magnetic Bead Kit (IDEXX Laboratories, Inc., One IDEXX Drive Westbrook, ME USA) and an automated instrument (Kingfisher Flex System magnetic beads processor, Thermo-Fisher Scientific, Waltham, MA, USA). In brief, nucleic acid purification was done as follows: 1) Sample (200 μL) and lysis solution (200 μL) were incubated for 15 min followed by 5 min incubation with the bead solution at room temperature (600 μL binding buffer +20 μL magnetic beads). 2) Magnetic beads were collected, washed with the solutions I, II, and 80 % ethanol (600 μL for 3 min each), and allowed to dry after the final wash (10 min at room temperature). 3) The nucleic acids were eluted from the magnetic beads having 100 μL of the elution buffer for 5 min at 24 °C room temperature.

PRRSV RT-rtPCR was performed using the RealPCR* PRRS Type 1 and Type 2 Multiplex RNA Mix and Master Mixes, (IDEXX Laboratories, Inc.). Purified RNA, Multiplex RNA mix and Master Mixes were combined in plates following the manufacturer's directions and then loaded onto a 7500 Fast Real-Time PCR System thermal cycler (Applied Biosystems©, Foster City, CA, USA). A standard ramp rate was used between temperatures in the thermal cycler. The cycling conditions used for the reaction are as follows: 1) One cycle at 50 °C for 15 min. 2) One cycle at 95 °C for 1 min. 3) Forty-five cycles of 95 °C for 15 s and 60 °C for 30 s.

The resultant amplification data was analyzed with the 'auto baseline' and 'auto Ct' functions. Samples with Cq values < 37 were considered PRRS-positive. One negative extraction control and positive and negative amplification controls were included in each run of the assay. Additionally, the IDEXX RealPCR* PRRS RT-rtPCR test included an internal sample control (ISC) to monitor sample quality, extraction, and amplification (i.e., endogenous, host specific genetic material). In brief, swine DNA within the sample is targeted by the ISC primers and probes and the reaction is validated by the amplification curve produced. All tests were performed at the Research and Development Laboratory of the Iowa State University College of Veterinary Medicine.

2.4. Data analysis

The effect of diluting a PRRSV-positive processing fluid sample with PRRSV-negative processing fluids (processing fluids from PRRSV-naïve litters) was modeled using 2-fold serial dilutions. The Cq value (28) of the PRRSV-positive stock solution was considered to represent a litter of 11 pigs containing one viremic piglet (López et al., 2020; Vilalta et al., 2018). Thus, each two-fold dilution theoretically doubled the number of litters (and pigs) represented in the processing fluids sample. After 8 dilutions, the processing fluids sample represented one viremic pig among 256 litters (~2857 piglets).

A univariate linear regression model was built to describe 'Cq values' (dependent variable) as a function of the 'number of pigs in the pool' (explanatory variable). The model allowed for the evaluation of the effect on Cq values of adding PRRSV-negative processing fluids to the original PRRSV-positive processing fluid sample. A probit regression model was fitted to calculate the probability of PRRSV RNA detection as a function of the number of PRRSV-negative pigs contributing to the processing fluid sample in each dilution. The probability of PRRSV RNA detection as the response variable was linked through a logistic distribution to the model. In both cases, the variable 'number of pigs in the pool' was Log2 transformed to linearize the relationship between the outcome and predictor variables and to facilitate visualization of data in a graph format. All analyses were performed using a commercial statistical software (SAS® Version 9.4, SAS® Institute, Inc., Cary, NC).

3. Results

This study evaluated the effect of pooling a PRRSV-positive processing fluids sample with known PRRSV-negative samples on the probability of detecting viral RNA by RT-rtPCR.

As expected, each two-fold dilution of the initial PRRSV-positive processing fluid sample with PRRSV-negative processing fluids increased the mean Cq by 1.37 (Fig. 1). There was an increasing dispersion of Cq values in higher dilutions. All replicates (n = 6) were RT-rtPCR-positive from the first until the fifth dilution, 33 % (2 of 6) of the replicates were RT-rtPCR-positive at the sixth dilution, and 17 % (1 of 6) of replicates were positive at the seventh dilution. All replicates were RT-rtPCR-negative at the eighth dilution (Table 1).

A graphical representation of the association between Cq values and the number of PRRSV-negative piglets based on linear regression analysis is shown in Fig. 2. Assuming the inclusion of one RT-rtPCR-positive

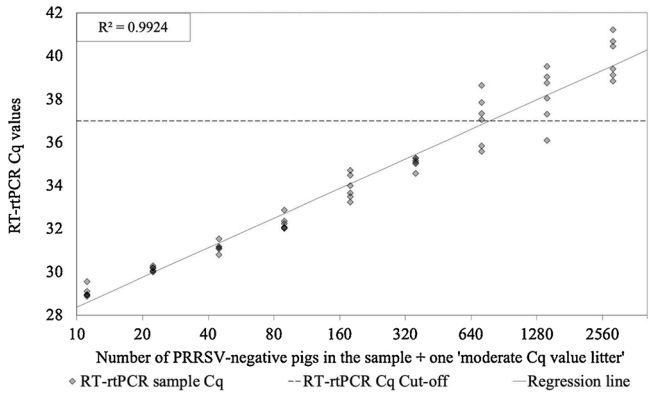

Fig. 1. Regression line for serially diluted processing fluid sample.

Table 1
Raw proportions for PRRSV RNA detection by RT-rtPCR in different levels of pooled processing fluids having only one viremic litter.

Dilution	N° of pigs in the pool	Frequency of detection[a] (%)
0 (undiluted)	11	6/6 (100 %)
1	22	6/6 (100 %)
2	45	6/6 (100 %)
3	89	6/6 (100 %)
4	179	6/6 (100 %)
5	357	6/6 (100 %)
6	714	2/6 (33.33 %)
7	1428	1/6 (16.66 %)
8	2857	0/6 (0.00 %)

[a] Proportion of PRRSV RT-rtPCR positive tests from 6 replicates.

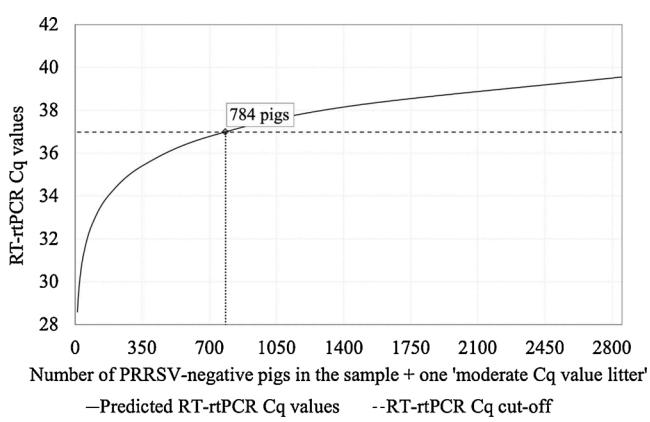

Fig. 2. Predicted Cq values as a function of the pool size based on the linear regression coefficients. Vertical dotted line indicates the last point where the predicted Cq value would yield a positive RT-rtPCR result and the associated number of pigs represented in the sample.

litter in the sample, a probit regression model (Fig. 3) estimated a probability of a positive RT-rtPCR result at 43 % in an aggregate sample containing 69 PRRSV-negative litters + one PRRSV-positive litter (784 piglets); 80 % in a sample containing 43 PRRSV-negative litters + one PRRSV-positive litter (492 piglets); and 95 % in a sample containing 28 PRRSV-negative litters + one PRRSV-positive litter (323 piglets), respectively.

4. Discussion

Responsible for the serological testing of WWII military recruits for syphilis, Dorfman's (1943) dilemma was to determine how many samples to pool to achieve maximum testing efficiency. For today's swine

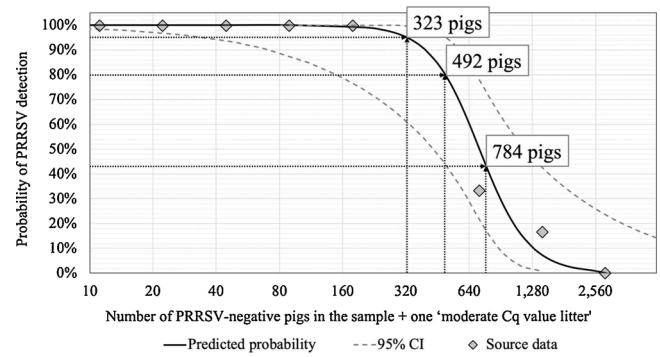

Fig. 3. Predicted probability for PRRSV RNA detection in pooled processing fluids with one' moderate Cq value litter.'.

industry, the appeal of moving away from individual pig samples to aggregate or pooled specimens is the reduction in the number of specimens to collect and tests to perform (Bilder and Tebbs, 2012). However, Dorfman's quandary (how many to pool?) remains pertinent and justifies continued efforts for achieving a better understanding of the capabilities and limitations of processing fluid-based sampling in PRRSV surveillance.

The objective of this study was to estimate the limits of PRRSV RNA detection in processing fluids and provide guidelines for the number of litters to include in processing fluid samples under field conditions. Models are commonly used to answer practical biological questions, and modeling was a practical approach to this question. To represent a conservative scenario, a PRRSV-positive processing fluid field sample with a "moderate" Cq value was used to model the effect of dilution on the detection of PRRSV RNA. That is, the probit regression model evaluated a 'worst-case scenario' in which processing fluids from one litter with a single viremic piglet were combined with processing fluids from known PRRSV-negative litters. The choice of selecting a sample with Cq value of 28 for the stock solution was supported by previous research showing that most PRRSV-positive litters (n = 22) having a single viremic pig had the median Cq value on processing fluid samples of 30.2 (López et al., 2020). Likewise, Vilalta et al. (2018), reported that 8 of 23 PRRSV-positive litters had a single viremic piglet with a median Cq value for processing fluid samples was 28.9.

The presented study helped to elucidate the limitations of PRRSV RNA detection in processing fluids. Also, it has taken further our understanding about the sensitivity of this diagnostic monitoring approach. Results of this study support the assumption that processing fluids samples originating from PRRSV-positive litters with Cq values below 28 and/or having more than a single viremic piglet, have a high probability of being deemed as positive for PRRSV RNA detection due to the high concentration of viral RNA in such sample. However, due to the aggregated nature of the processing fluids sampling approach, Cq values from PF samples can not be used to estimate prevalence (i.e., number of pigs, or number of litters testing PRRSV-positive).

In conclusion, the regression analysis results for the serially diluted processing fluid sample suggest that limiting processing fluid sampling to ≤ 30 litters (~320 piglets) is expected to provide a 95 % confidence in detecting PRRSV RNA even when prevalence within piglets is near-zero. Because it allows for testing more pigs more frequently, processing fluid-based surveillance offers an economic advantage over individual pig sampling approaches. Thus, the knowledge generated in this study regarding processing fluids pooling alternatives and its limits extends the reach of PRRSV surveillance capabilities in breeding herds in considerable measure by further reducing the cost associated with diagnostic testing.

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Declaration of Competing Interest

The authors declare no conflicts of interest with respect to the conduct, authorship, and/or publication of this study. Co-author JZ has served as a consultant to IDEXX Laboratories, Inc. on areas of diagnostic medicine independent of this research. The terms of the consulting arrangement have been reviewed and approved by Iowa State University in accordance with its conflict of interest policies. Co-author JA is employed by Zoetis Inc.

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